Enterprise EIGRP Network Implementation Project

Professional Configuration Documentation

This document contains the complete configuration details for the Enterprise EIGRP WAN Topology Project. All configurations are preserved exactly as implemented in the topology, including LAN configuration, WAN addressing, EIGRP routing, ISP connectivity, and VRRP redundancy implementation.

Configuration Component	Status
LAN Addressing	Verified
WAN Addressing	Verified
HQ ↔ ISP Connectivity	Verified
Branch ↔ ISP Connectivity	Verified
EIGRP Configuration	Verified
EIGRP Neighbor Formation	Verified
Branch-to-Branch Connectivity	Verified
VRRP Redundancy	Verified

STEP 1 — LAN Addressing & LAN Connectivity

```
STEP 1> Assign the IP Addresses respectively to the HQ-Region LAN and the SUB-BRANCHES Location and then test fi

                           HQ-BANGALORE:

PC1:
ip 10.0.1.1/24 10.0.1.40

PC2:
ip 10.0.1.3/24 10.0.1.40

SERVER:
ip 10.0.1.2/24 10.0.1.50


HQ_ROUTER_1:

en
conf t
line console 0
logg syn
no exec-timeout 0 0
ho HQ_ROUTER_1

int f0/0
ip add 10.0.1.40 255.255.255.0
no sh
do wr
```

```
do sh ip int br
```

HQ_ROUTER_2:

```
en
conf t
line console 0
logg syn
no exec-timeout 0 0
ho HQ_ROUTER_2

int f0/0
ip add 10.0.1.50 255.255.255.0
no sh
do wr
do sh ip int br
```

BRANCH-1 KOLKATA

PC1:

```
ip 192.168.1.1/24 192.168.1.10
save
```

PC2:

```
ip 192.168.1.2/24 192.168.1.10
save
```

PC3:

```
ip 192.168.1.3/24 192.168.1.20
save
```

R1:

```
en
conf t
line console 0
logg syn
no exec-timeout 0 0
ho BRANCH_1_R1

int f2/0
ip add 192.168.1.10 255.255.255.0
no sh
do wr
do sh ip int br
```

R2:

```
en
conf t
line console 0
logg syn
no exec-timeout 0 0
ho BRANCH_1_R2

int f2/0
ip add 192.168.1.20 255.255.255.0
no sh
do wr
do sh ip int br
```

■■■ BRANCH-2 ASSAM

PC1:

```
ip 192.168.2.1/24 192.168.2.10
save
```

PC2:

```
ip 192.168.2.2/24 192.168.2.10
save

PC3:
ip 192.168.2.3/24 192.168.2.20
save


R1:

en
conf t
line console 0
logg syn
no exec-timeout 0 0
ho BRANCH_2_R1

int f2/0
ip add 192.168.2.10 255.255.255.0
no sh
do wr
do sh ip int br


R2:

en
conf t
line console 0
logg syn
no exec-timeout 0 0
ho BRANCH_2_R2

int f2/0
ip add 192.168.2.20 255.255.255.0
no sh
do wr
do sh ip int br
```

■■■ BRANCH-3 DELHI

```
PC1:
ip 192.168.3.1/24 192.168.3.10
save

PC2:
ip 192.168.3.2/24 192.168.3.10
save

PC3:
ip 192.168.3.3/24 192.168.3.20
save


R1:

en
conf t
line console 0
logg syn
no exec-timeout 0 0
ho BRANCH_3_R1

int f2/0
ip add 192.168.3.10 255.255.255.0
no sh
do wr
do sh ip int br
R2:
en
```

```
conf t
line console 0
logg syn
no exec-timeout 0 0
ho BRANCH_3_R2

int f2/0
ip add 192.168.3.20 255.255.255.0
no sh
do wr
do sh ip int br

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```

STEP 2 — WAN Addressing Between Branches and ISPs

```
STEP 2> Assign IP Addresses between ISP and BRANCH network and test the WAN connectivity


■■BRANCH-1 KOLKATA <-> ISP_1 & ISP_2

BRANCH_1_R1 <-> ISP_1

R1:

int f0/0
ip add 172.16.13.10 255.255.255.0
no sh
do wr
do sh ip int br


ISP_1

en
conf t
ho ISP_1
line console 0
logg syn
no exec-timeout 0 0

int f2/0
ip add 172.16.13.1 255.255.255.0
no sh
do wr
do sh ip int br


BRANCH_1_R1 <-> ISP_2

R1:

int f1/0
ip add 172.16.14.10 255.255.255.0
no sh
do wr
do sh ip int br


ISP_2

en
conf t
ho ISP_2
line console 0
logg syn
no exec-timeout 0 0
```

```
int f2/0
ip add 172.16.14.1 255.255.255.0
no sh
do wr
do sh ip int br
```

BRANCH_1_R2 <-> ISP_1

R2:

```
int f0/0
ip add 172.16.15.20 255.255.255.0
no sh
do wr
do sh ip int br
```

ISP_1

```
en
conf t
line console 0
logg syn
no exec-timeout 0 0

int f3/0
ip add 172.16.15.1 255.255.255.0
no sh
do wr
do sh ip int br
```

BRANCH_1_R2 <-> ISP_2

R2:

```
int f1/0
ip add 172.16.16.20 255.255.255.0
no sh
do wr
do sh ip int br
```

ISP_2

```
en
conf t
line console 0
logg syn
no exec-timeout 0 0

int f3/0
ip add 172.16.16.1 255.255.255.0
no sh
do wr
do sh ip int br
```

■■BRANCH-2 ASSAM <-> ISP_3 & ISP_4

BRANCH_2_R1 <-> ISP_3

R1:

```
int f0/0
ip add 172.16.17.10 255.255.255.0
```

```
no sh
do wr
do sh ip int br
```

ISP_3

```
en
conf t
ho ISP_3
line console 0
logg syn
no exec-timeout 0 0

int f2/0
ip add 172.16.17.1 255.255.255.0
no sh
do wr
do sh ip int br
```

BRANCH_2_R1 <-> ISP_4

R1:

```
int f1/0
ip add 172.16.18.10 255.255.255.0
no sh
do wr
do sh ip int br
```

ISP_4

```
en
conf t
ho ISP_4
line console 0
logg syn
no exec-timeout 0 0

int f2/0
ip add 172.16.18.1 255.255.255.0
no sh
do wr
do sh ip int br
```

BRANCH_2_R2 <-> ISP_3

R2:

```
int f0/0
ip add 172.16.19.20 255.255.255.0
no sh
do wr
do sh ip int br
```

ISP_3

```
int f3/0
ip add 172.16.19.1 255.255.255.0
no sh
do wr
do sh ip int br
```

BRANCH_2_R2 <-> ISP_4

R2:

```
int f1/0
ip add 172.16.20.20 255.255.255.0
no sh
do wr
do sh ip int br
```

ISP_4

```
int f3/0
ip add 172.16.20.1 255.255.255.0
no sh
do wr
do sh ip int br
```

■■ BRANCH-3 DELHI <-> ISP_5 & ISP_6

BRANCH_3_R1 <-> ISP_5

R1:

```
int f0/0
ip add 172.16.21.10 255.255.255.0
no sh
do wr
do sh ip int br
```

ISP_5

```
en
conf t
ho ISP_5
line console 0
logg syn
no exec-timeout 0 0

int f2/0
ip add 172.16.21.1 255.255.255.0
no sh
do wr
do sh ip int br
```

BRANCH_3_R1 <-> ISP_6

R1:

```
int f1/0
ip add 172.16.22.10 255.255.255.0
no sh
do wr
do sh ip int br
```

ISP_6

```
en
conf t
ho ISP_6
line console 0
logg syn
no exec-timeout 0 0

int f2/0
```

```
ip add 172.16.22.1 255.255.255.0
no sh
do wr
do sh ip int br



BRANCH_3_R2 <-> ISP_5

R2:

int f0/0
ip add 172.16.23.20 255.255.255.0
no sh
do wr
do sh ip int br


ISP_5

en
conf t
ho ISP_5
line console 0
logg syn
no exec-timeout 0 0

int f3/0
ip add 172.16.23.1 255.255.255.0
no sh
do wr
do sh ip int br



BRANCH_3_R2 <-> ISP_6

R3:

int f1/0
ip add 172.16.24.20 255.255.255.0
no sh
do wr
do sh ip int br


ISP_6

en
conf t
ho ISP_6
line console 0
logg syn
no exec-timeout 0 0

int f3/0
ip add 172.16.24.1 255.255.255.0
no sh
do wr
do sh ip int br


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```

STEP 3 — WAN Addressing Between HQ and ISPs

```
STEP 3> Assign IP Addresses between ISP and HEADQUARTER(Bangalore) network and test the WAN connectivity
```

```
HQ_ROUTER_1 <-> ISP_1, ISP_2, ISP_3, ISP_4, ISP_5 and ISP_6

R1:

en
conf t
int f1/0
ip add 172.16.1.40 255.255.255.0
no sh
do wr


ISP_1:

en
conf t
int f0/0
ip add 172.16.1.1 255.255.255.0
no sh
do wr


R1:

int f2/0
ip add 172.16.2.40 255.255.255.0
no sh
do wr


ISP_2:

en
conf t
int f0/0
ip add 172.16.2.1 255.255.255.0
no sh
do wr


R1:

int f3/0
ip add 172.16.3.40 255.255.255.0
no sh
do wr


ISP_3:

en
conf t
int f0/0
ip add 172.16.3.1 255.255.255.0
no sh
do wr


R1:

int f4/0
ip add 172.16.4.40 255.255.255.0
no sh
do wr


ISP_4:

en
conf t
int f0/0
```

```
ip add 172.16.4.1 255.255.255.0
no sh
do wr


R1:

int f5/0
ip add 172.16.5.40 255.255.255.0
no sh
do wr


ISP_5:

en
conf t
int f0/0
ip add 172.16.5.1 255.255.255.0
no sh
do wr



R1:

int f6/0
ip add 172.16.6.40 255.255.255.0
no sh
do wr


ISP_6:

en
conf t
int f0/0
ip add 172.16.6.1 255.255.255.0
no sh
do wr



HQ_ROUTER_2 <-> ISP_1, ISP_2, ISP_3, ISP_4, ISP_5 and ISP_6


R2:

en
conf t
int f1/0
ip add 172.16.7.50 255.255.255.0
no sh
do wr


ISP_1:

int f1/0
ip add 172.16.7.1 255.255.255.0
no sh
do wr


R2:

int f2/0
ip add 172.16.8.50 255.255.255.0
no sh
do wr
ISP_2:
```

```
int f1/0
ip add 172.16.8.1 255.255.255.0
no sh
do wr


R2:

int f3/0
ip add 172.16.9.50 255.255.255.0
no sh
do wr


ISP_3:

int f1/0
ip add 172.16.9.1 255.255.255.0
no sh
do wr


R2:

int f4/0
ip add 172.16.10.50 255.255.255.0
no sh
do wr


ISP_4:

int f1/0
ip add 172.16.10.1 255.255.255.0
no sh
do wr


R2:

int f5/0
ip add 172.16.11.50 255.255.255.0
no sh
do wr


ISP_5:

int f1/0
ip add 172.16.11.1 255.255.255.0
no sh
do wr



R2:

int f6/0
ip add 172.16.12.50 255.255.255.0
no sh
do wr
do sh ip int br


ISP_6:

int f1/0
ip add 172.16.12.1 255.255.255.0
no sh
do wr
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```

STEP 4 — EIGRP Configuration on HQ Routers

```
STEP 4> Configure EIGRP on HQ_ROUTERS

HQ_ROUTER_1:

en
conf t
router eigrp 100
no auto-summary

network 10.0.1.0
network 172.16.1.0
network 172.16.2.0
network 172.16.3.0
network 172.16.4.0
network 172.16.5.0
network 172.16.6.0
do wr
exit


HQ_ROUTER_2:

en
conf t
router eigrp 100
no auto-summary

network 10.0.1.0
network 172.16.7.0
network 172.16.8.0
network 172.16.9.0
network 172.16.10.0
network 172.16.11.0
network 172.16.12.0
do wr
exit

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```

STEP 5 — EIGRP Configuration on Branch Routers

```
STEP 5> Configure EIGRP on the BRANCH_ROUTERS

BRANCH_1_R1:

en
conf t
router eigrp 100
no auto-summary

network 192.168.1.0
network 172.16.13.0
network 172.16.14.0
do wr
exit


BRANCH_1_R2:

en
conf t
```

```
router eigrp 100
no auto-summary

network 192.168.1.0
network 172.16.15.0
network 172.16.16.0
do wr
exit
```

BRANCH_2_R1:

```
en
conf t
router eigrp 100
no auto-summary

network 192.168.2.0
network 172.16.17.0
network 172.16.18.0
do wr
exit
```

BRANCH_2_R2:

```
en
conf t
router eigrp 100
no auto-summary

network 192.168.2.0
network 172.16.19.0
network 172.16.20.0
do wr
exit
```

BRANCH_3_R1:

```
en
conf t
router eigrp 100
no auto-summary

network 192.168.3.0
network 172.16.21.0
network 172.16.22.0
do wr
exit
```

BRANCH_3_R2:

```
en
conf t
router eigrp 100
no auto-summary

network 192.168.3.0
network 172.16.23.0
network 172.16.24.0
do wr
exit
```

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STEP 6 — EIGRP Configuration on ISP Routers

```
STEP 6> Configure EIGRP on the ISP_ROUTERS



ISP_1:

en
conf t
router eigrp 100
no auto-summary

network 172.16.13.0
network 172.16.15.0
network 172.16.1.0
network 172.16.7.0
do wr
exit



ISP_2:

en
conf t
router eigrp 100
no auto-summary

network 172.16.14.0
network 172.16.16.0
network 172.16.2.0
network 172.16.8.0
do wr
exit



ISP_3:

en
conf t
router eigrp 100
no auto-summary

network 172.16.17.0
network 172.16.19.0
network 172.16.3.0
network 172.16.9.0
do wr
exit



ISP_4:

en
conf t
router eigrp 100
no auto-summary

network 172.16.18.0
network 172.16.20.0
network 172.16.4.0
network 172.16.10.0
do wr
exit
ISP_5:

en
```

```
conf t
router eigrp 100
no auto-summary

network 172.16.21.0
network 172.16.23.0
network 172.16.5.0
network 172.16.11.0
do wr
exit



ISP_6:

en
conf t
router eigrp 100
no auto-summary

network 172.16.22.0
network 172.16.24.0
network 172.16.6.0
network 172.16.12.0
do wr
exit

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```

STEP 7 — VRRP Configuration

```
STEP 6> Configure VRRP on HQ and BRANCH Routers


HQ VRRP Configuration

HQ_R1(MASTER):

interface f0/0
vrrp 1 ip 10.0.1.5
vrrp 1 priority 120
vrrp 1 preempt
do wr
exit

HQ_R2(BACKUP):

interface f0/0
vrrp 1 ip 10.0.1.5
vrrp 1 priority 100
vrrp 1 preempt
do wr
exit


BRANCH_1 VRRP Configuration

BR_1_R1(MASTER):

interface f2/0
vrrp 1 ip 192.168.1.100
vrrp 1 priority 120
vrrp 1 preempt
do wr
exit

BR_1_R2(BACKUP):
```

```
interface f2/0
vrrp 1 ip 192.168.1.100
vrrp 1 priority 100
vrrp 1 preempt
do wr
exit



BR_2_R1(MASTER):

interface f2/0
vrrp 1 ip 192.168.2.100
vrrp 1 priority 120
vrrp 1 preempt
do wr exit

BR_2_R2(BACKUP):

interface f2/0
vrrp 1 ip 192.168.2.100
vrrp 1 priority 100
vrrp 1 preempt
do wr
exit



BR_3_R1(MASTER):

interface f2/0
vrrp 1 ip 192.168.3.100
vrrp 1 priority 120
vrrp 1 preempt
do wr
exit

BR_3_R2(BACKUP):

interface f2/0
vrrp 1 ip 192.168.3.100
vrrp 1 priority 100
vrrp 1 preempt
do wr
exit


*** CHANGE THE DEFAULT GATEWAY OF THE RESPECTIVE PCs TO THE VRRP ADDRESS ***
```

Project Status:
The enterprise WAN topology has been successfully implemented with:

• Multi-branch enterprise connectivity
• Dynamic routing using EIGRP
• Equal-Cost Load Balancing (ECMP)
• Automatic WAN failover
• VRRP-based first-hop redundancy
• End-to-end branch communication